

Submission CALC1-QUIZ-SUB01

Question CALC1-QUIZ-Q01

scratch: show step

1. $f'(x) = 2x(2x-1) + (x^2+5)^2$

2. $f'(1) = 2(1) + (6)^2 = 12$

=> FINAL: 12 (check units/interpretation)

Question CALC1-QUIZ-Q02

scratch: show step

1. $\int 5x \, dx = (5/2)x^2$

2. $[5/2 x^2]_0^2 = 10$

=> FINAL: 10 (check units/interpretation)

Question CALC1-QUIZ-Q03

scratch: show step

1. $2w+2l=20 \Rightarrow l=10-w$

2. $A(w) = w(10-w)$; at $w=7$, $A=21$

3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.

=> FINAL: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (check units/interpretation)

Question CALC1-QUIZ-Q04

scratch: show step

1. $x^2-36=(x-6)(x+6)$

2. cancel $x-a$ only after identifying $x \neq a$

3. $\lim_{x \rightarrow 6} \frac{x^2-36}{x-6} = 12$

=> FINAL: 12 (check units/interpretation)

Question CALC1-QUIZ-Q05

scratch: show step

1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).

2. A sign change in f' from + to – indicates a local maximum.

=> FINAL: critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum (check units/interpretation)